

Resonance-Avoidance Speed Control for Bridges: A Self-Optimizing Traffic–Structure Interaction Framework

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Abstract— Bridge resonance induced by moving vehicular loads often leads to excessive vibration, accelerated fatigue damage, and a reduced service life. While conventional mitigation techniques rely on costly and inflexible structural modifications like increased stiffness or damping, this research proposes a *novel Resonance-Avoidance Speed Control (RASC) framework*. By treating traffic speed as a controllable system variable, the framework creates a *self-optimizing traffic–structure interaction system* that eliminates the need for physical retrofitting.

The methodology integrates structural dynamics, moving load theory, and control logic to dynamically regulate vehicle speeds. It identifies "*resonance zones*"—specific speed intervals where the fundamental excitation frequency of moving loads coincides with the bridge's natural frequencies—and enforces speed adjustments to keep traffic within safe operating bands. Validated through numerical simulations, this system-level optimization is expected to significantly reduce dynamic amplification and decrease vibration amplitudes by 20–50%. Ultimately, this scalable and non-intrusive approach offers a low-cost alternative to traditional structural modifications, successfully improving fatigue life and introducing a new paradigm for smart, resilient infrastructure management

Keywords— Bridge Resonance, Resonance-Avoidance Speed Control (RASC), Traffic–Structure Interaction, Moving Load Theory, Dynamic Amplification Factor (DAF), Fatigue Damage Mitigation, Non-Intrusive Control

I. INTRODUCTION

The continuous movement of vehicular traffic across bridges introduces dynamic loads that can significantly impact structural integrity. When the fundamental excitation frequency of these moving loads aligns with the natural frequencies of the bridge, it induces resonance. This phenomenon leads to **excessive vibration, accelerated fatigue damage, and a progressively reduced service life** for the infrastructure.

Historically, engineers have relied on conventional mitigation techniques to combat resonance, primarily through physical structural modifications such as increasing the bridge's stiffness or adding mechanical damping. While effective, these traditional retrofitting methods are **notoriously costly, inflexible, and highly intrusive**.

To address these critical limitations, this research introduces a novel **Resonance-Avoidance Speed Control (RASC) framework**,

designed as a self-optimizing traffic–structure interaction system. Rather than physically altering the bridge, the fundamental philosophy of this framework is to **treat traffic speed as a controllable system variable**. Because the excitation frequency of a moving vehicle is directly tied to its speed and the bridge's span, adjusting vehicle speeds can actively prevent resonance.

The proposed methodology integrates structural dynamics, moving load theory, and advanced control logic. By evaluating the dynamic amplification factor (DAF) across various speeds, the system identifies "**resonance zones**"—**specific speed intervals where the structural response exceeds safe thresholds**. Once these zones are mapped, the RASC framework dynamically regulates allowable vehicle speeds, enforcing speed adjustments to ensure traffic remains within safe operating bands while avoiding restricted resonance bands.

By redefining traffic as a controllable excitation source, this research establishes a **system-level co-optimization framework that couples traffic behavior with structural health**. This scalable, non-intrusive approach is compatible with existing traffic control infrastructure and offers a low-cost alternative to traditional retrofitting. Ultimately, the RASC framework aims to **reduce vibration amplitudes by 20–50%** and significantly improve fatigue life, introducing a new paradigm in smart, resilient infrastructure design.

II. METHODOLOGY

The Resonance-Avoidance Speed Control (RASC) framework is developed as a structured approach to study and mitigate resonance in bridges subjected to moving vehicular loads. Rather than relying on structural modifications, the proposed method focuses on controlling the source of excitation—vehicle speed. The methodology combines structural dynamics, moving load analysis, and a control-based perspective to identify and avoid resonance conditions.

A. System Overview

The overall approach follows a step-by-step process. First, the bridge is modeled and its dynamic characteristics are obtained through modal analysis. This provides the natural frequencies of the structure, which are essential for identifying resonance conditions. Next, moving load simulations are carried out by varying vehicle speeds across a defined range. Since vehicle speed directly influences excitation frequency, this step allows the observation of how the bridge responds under different operating conditions. By comparing the excitation frequency with the natural frequencies, speed ranges that lead to resonance are identified. These ranges,

referred to as *resonance zones*, correspond to conditions where the structural response is significantly amplified. Finally, a control strategy is formulated to ensure that vehicles operate outside these critical speed intervals.

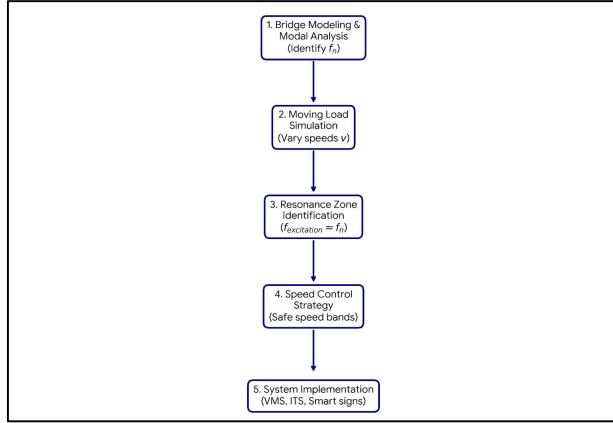


Fig 1. Block diagram of the RASC framework.

B. Dynamic Modeling of Bridge

The bridge is represented as a multi-degree-of-freedom dynamic system using finite element modeling. Its response to external loading is governed by the standard equation of motion:

$$M\ddot{x}(t) + C\dot{x}(t) + Kx(t) = F(t)$$

where the matrices M , C , and K represent the mass, damping, and stiffness properties of the structure. The term $F(t)$ corresponds to the time-varying load introduced by moving vehicles, while $x(t)$ describes the structural response.

A detailed finite element model is developed to capture the geometry and material properties of the bridge components, including the deck, girders, piers, and diaphragms. This allows for an accurate estimation of natural frequencies and mode shapes, which form the basis for subsequent resonance analysis.

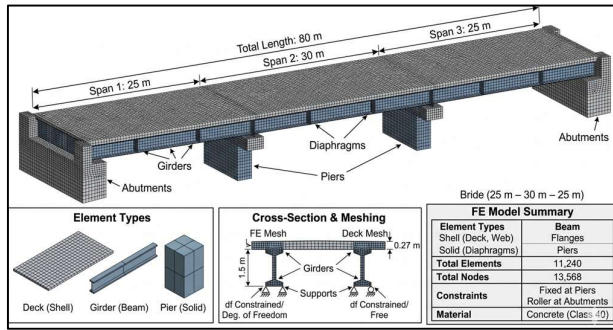


Fig 2. Finite element model of the bridge.

C. Moving Load Modeling

Vehicular loading is modeled using a simplified two-axle system moving across the bridge deck. Each axle applies a constant load, and the spacing between axles is maintained as a fixed parameter. Although simplified, this model effectively captures the essential characteristics required for studying resonance phenomena.

The frequency of excitation generated by the moving load is directly related to vehicle speed and axle spacing, and can be expressed as:

$$f_{excitation} = \frac{v}{d}$$

where v is the vehicle speed and d is the axle spacing. This relationship highlights that speed plays a critical role in determining whether resonance conditions will occur.

By varying the speed across a suitable range, different excitation frequencies are produced, allowing the identification of conditions where the structural response becomes critical.

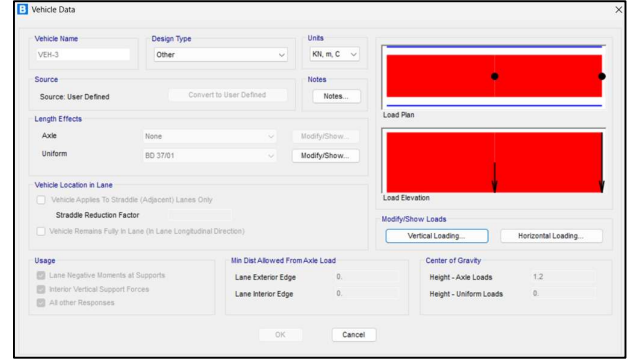


Fig 3. Schematic representation of moving load and axle spacing.

D. Resonance Zone Identification

To identify resonance zones, the bridge response is evaluated for different vehicle speeds. Parameters such as displacement, acceleration, and Dynamic Amplification Factor (DAF) are monitored for each simulation case.

As the vehicle speed varies, the structural response exhibits noticeable peaks at certain speeds. These peaks indicate resonance conditions, where the excitation frequency approaches one of the natural frequencies of the bridge. The corresponding speed intervals are defined as resonance zones.

A response-versus-speed plot provides a clear visualization of these regions, making it easier to identify critical speed ranges that should be avoided.

E. Development of Speed Control Strategy

Once the resonance zones are identified, a speed control strategy is developed to prevent vehicles from operating within these critical ranges. Instead of allowing unrestricted speeds, the framework defines allowable speed bands that lie outside resonance-prone regions.

The process involves determining natural frequencies, computing the corresponding critical speeds, and establishing safe operating intervals. Traffic speeds can then be regulated using conventional or smart traffic management systems to ensure that vehicles remain within these safe limits.

This approach effectively minimizes the likelihood of resonance without requiring any physical changes to the bridge structure.

F. Implementation Framework

The methodology is implemented using a simulation-based approach. The bridge model is developed in finite element software such as SAP2000 or CSI Bridge, where modal analysis and moving load simulations are performed.

The performance of the RASC framework is evaluated by comparing structural responses under two conditions: normal traffic

operation and controlled speed operation. The comparison highlights the effectiveness of the proposed strategy in reducing vibration levels and improving structural performance.

III. MATHEMATICAL FORMULATION

To establish a quantitative understanding of resonance behavior and its impact on structural performance, a mathematical framework is developed. This formulation links vehicle speed, excitation frequency, structural response, and fatigue life, forming the analytical basis of the proposed RASC strategy.

A. Resonance Condition

Resonance in bridge structures occurs when the excitation frequency induced by moving vehicular loads approaches or matches the natural frequency of the bridge. For a simplified moving load model, the excitation frequency depends on the vehicle speed and axle spacing.

$$f = \frac{v}{d}$$

Where,

f is the excitation frequency,

v is the vehicle speed,

d is the distance between consecutive axles.

Resonance occurs when this excitation frequency becomes equal (or very close) to the natural frequency f_n of the bridge. Under such conditions, even small periodic forces can produce large-amplitude vibrations due to dynamic amplification. This relationship forms the fundamental criterion for identifying resonance conditions in the proposed framework.

B. Critical Speed

The critical speed is defined as the vehicle speed at which resonance occurs. It can be obtained by equating the excitation frequency to the natural frequency of the bridge.

$$v = f_n \cdot d$$

Where,

f_n represents the natural frequency of the bridge corresponding to a particular vibration mode.

This expression shows that for each natural frequency; there exists a corresponding critical speed at which resonance may occur. In practical scenarios, multiple modes of vibration exist, resulting in multiple critical speeds. These speeds define the resonance-prone regions that must be avoided through appropriate control measures.

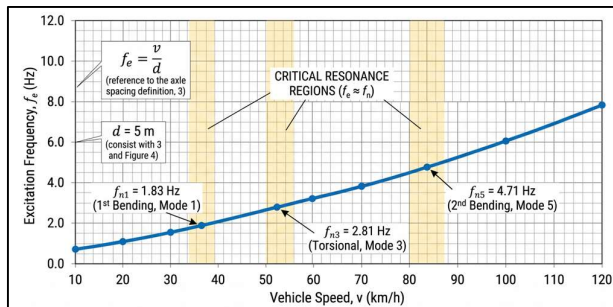


Fig 5. Linear relationship between vehicle speed (v) and excitation frequency (f_e) for a fixed axle spacing $d=5$ m. Shaded bands highlight speeds where f_e matches the bridge's natural frequencies (f_n) as defined in Figures 0, 3, and 4

Fig 5. Graph: vehicle speed and excitation frequency.

C. Fatigue Life Relation

Repeated dynamic loading due to vehicular movement leads to fatigue damage in bridge components. The fatigue life of a structure is strongly influenced by the amplitude of cyclic stress experienced during operation.

The relationship between fatigue life and stress amplitude can be expressed as:

$$N \propto \frac{1}{\sigma^m}$$

Where,

N is the number of cycles to failure (fatigue life),

σ is the stress amplitude,

m is a material-dependent constant.

This relationship indicates that fatigue life decreases rapidly with an increase in stress amplitude. Since resonance significantly amplifies structural response, it leads to higher stress levels and consequently accelerates fatigue damage.

D. Life Reduction Under Resonance

Under resonance conditions, the stress amplitude increases due to dynamic amplification. Let the stress under normal conditions be σ_0 , and the stress under resonance be expressed as:

$$\sigma = \alpha \sigma_0$$

where $\alpha > 1$ represents the amplification factor.

Substituting this into the fatigue life relation:

$$N_{res} = \frac{N_0}{\alpha^m}$$

Where,

N_0 is the original fatigue life,

N_{res} is the reduced fatigue life under resonance conditions.

For the present study, the stress amplification factor is observed to be approximately $\alpha = 1.31$. For typical material behavior where $m \approx 3$, the fatigue life reduces significantly:

$$N_{res} \approx \frac{N_0}{(1.31)^3} \approx 0.44 N_0$$

This indicates that the fatigue life reduces to nearly 44% of its original value when the structure operates under resonance conditions.

Such a drastic reduction highlights the importance of avoiding resonance through appropriate control strategies, as even moderate increases in stress can lead to substantial decreases in service life.

IV. BRIDGE MODELING AND SIMULATION

This section presents the modeling and simulation framework used to analyze the dynamic behavior of the bridge under moving vehicular loads. A continuous girder bridge is selected as the case study, and its structural response is evaluated using finite element analysis. The modeling process includes geometric definition, 3D visualization, load application, and modal analysis.

A. Bridge Description

A continuous girder bridge configuration is adopted for the study due to its widespread use in medium-span highway bridges and its susceptibility to dynamic effects under moving loads. The bridge consists of three spans with a symmetric layout and multiple longitudinal girders supporting the deck.

The geometric and structural parameters used in the model are summarized in Table I.

Parameter	Value
Span Configuration	25 m – 30 m – 25 m
Total Length	80 m
Deck Width	10 m
Deck Thickness	0.27 m
Number of Girders	4
Girder Depth	1.5 m
Girder Spacing	2.5 m
Diaphragm Spacing	5 m
Pier Height	7 m

Table I. Bridge Design Parameters.

The bridge is designed as a two-lane structure with three spans supported by piers and abutments. The superstructure consists of precast I-girders, which provide efficient load transfer and structural stability.

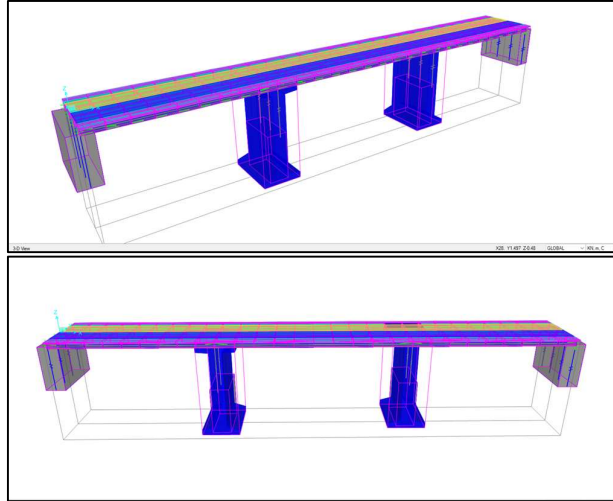


Fig 6. 3D representation of the continuous girder bridge.

B. 3D Modeling

To accurately represent the geometry and structural configuration, a detailed three-dimensional model of the bridge is developed. The modeling process begins with the creation of the bridge structure in a 3D environment using Blender. This includes the deck, girders, piers, and diaphragms, ensuring proper alignment and realistic proportions.

The 3D model is then exported in STL format and imported into CAD software such as FreeCAD to generate technical drawings and ensure dimensional accuracy. These models are further utilized in finite element analysis software for structural simulation.

The use of 3D modeling enhances visualization and ensures that geometric complexities are accurately captured, leading to more reliable simulation results.

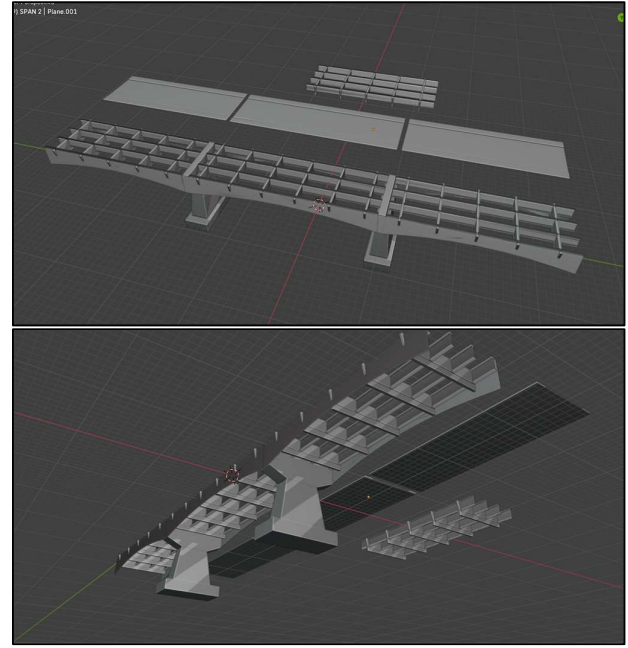


Fig 7. 3D bridge model.

C. Load Definition

The bridge is analyzed under moving vehicular loads using a simplified two-axle vehicle model. Each axle is assigned a concentrated load, and the spacing between the axles is maintained as a fixed parameter to control excitation frequency.

The load characteristics used in the study are summarized in Table II.

Parameter	Value
Number of Axles	2
Load per Axle	80 kN
Axle Spacing	4 m
Total Load	160 kN

Table II. Vehicle Load Parameters

In addition to the user-defined vehicle model, standard loading references such as IRC Class A and Class 70R are considered for validation. However, the custom vehicle model is preferred for resonance analysis, as it provides better control over axle spacing and excitation frequency.

The moving load is simulated across the bridge at varying speeds to evaluate the dynamic response under different excitation conditions.

Loads		Uniform Load Scale Factor			Axle Load Scale Factor			
Load Length Type	Minimum Distance	Maximum Distance	Uniform Load	Uniform Width Type	Uniform Width	Axle Load	Axle Width Type	Axle Width
Fixed Length	1	80	Zero Width		1	One Point		
Fixed Length	1	80	Zero Width		1	One Point		
Fixed Length	1	80	Zero Width		2	One Point		

Fig 8. Schematic of vehicle load model.

D. Modal Analysis Setup

Modal analysis is performed to determine the natural frequencies and corresponding mode shapes of the bridge. This step is critical for identifying resonance conditions, as it defines the inherent vibration characteristics of the structure.

The finite element model is developed in structural analysis software such as SAP2000 or CSI Bridge, where appropriate boundary conditions and material properties are assigned. The bridge is modeled with realistic support conditions at piers and abutments to accurately capture structural behavior.

Eigenvalue analysis is carried out to extract the natural frequencies of the system. Multiple modes are considered, as resonance can occur at different frequencies corresponding to different mode shapes.

The fundamental frequency obtained from the analysis serves as the primary parameter for calculating critical speeds and identifying resonance zones.

EIGEN MODAL ANALYSIS				14:17:21
CASE: NATFREQ				
USING STIFFNESS AT ZERO (UNSTRESSED) INITIAL CONDITIONS				
NUMBER OF STIFFNESS DEGREES OF FREEDOM	=	11250		
NUMBER OF MASS DEGREES OF FREEDOM	=	6453		
MAXIMUM NUMBER OF EIGEN MODES SOUGHT	=	20		
MINIMUM NUMBER OF EIGEN MODES SOUGHT	=	1		
NUMBER OF RESIDUAL-MASS MODES SOUGHT	=	0		
NUMBER OF SUBSPACE VECTORS USED	=	24		
RELATIVE CONVERGENCE TOLERANCE	=	1.00E-09		
FREQUENCY SHIFT (CENTER) (CYC/TIME)	=	.000000		
FREQUENCY CUTOFF (RADIUS) (CYC/TIME)	=	-INFINITY		
ALLOW AUTOMATIC FREQUENCY SHIFTING	=	YES		
Original stiffness at shift : EV= 0.0000000E+00, f= .000000, T= -INFINITY				
Number of eigenvalues below shift = 0				
Found mode	1 of	20:	EV= 3.4313659E-07, f= 9.32E-05, T= 10726.211	
Found mode	2 of	20:	EV= 1.5442194E+02, f= 1.975045, T= 0.505294	
Found mode	3 of	20:	EV= 3.1507489E+02, f= 2.825056, T= 0.353975	
Found mode	4 of	20:	EV= 4.5250628E+02, f= 3.385575, T= 0.295371	
Found mode	5 of	20:	EV= 4.5656523E+02, f= 3.400725, T= 0.294055	
Found mode	6 of	20:	EV= 7.3207465E+02, f= 4.306236, T= 0.232221	
Found mode	7 of	20:	EV= 8.6852513E+02, f= 4.690686, T= 0.213188	
Found mode	8 of	20:	EV= 1.3035569E+03, f= 6.943894, T= 0.144011	
Found mode	9 of	20:	EV= 2.846542E+03, f= 8.491858, T= 0.117760	
Found mode	10 of	20:	EV= 2.846542E+03, f= 8.491858, T= 0.117760	
Found mode	11 of	20:	EV= 3.1505032E+03, f= 8.933262, T= 0.111941	
Found mode	12 of	20:	EV= 3.2796457E+03, f= 9.114516, T= 0.109715	
Found mode	13 of	20:	EV= 3.6175472E+03, f= 9.572541, T= 0.104465	
Found mode	14 of	20:	EV= 4.2635259E+03, f= 10.392135, T= 0.096227	
Found mode	15 of	20:	EV= 5.1251955E+03, f= 11.398429, T= 0.087731	
Found mode	16 of	20:	EV= 5.3636922E+03, f= 11.656057, T= 0.085792	
Found mode	17 of	20:	EV= 5.9887347E+03, f= 12.316512, T= 0.081192	
Found mode	18 of	20:	EV= 6.3838086E+03, f= 12.716279, T= 0.078639	
Found mode	19 of	20:	EV= 6.9735905E+03, f= 13.291096, T= 0.075238	
Found mode	20 of	20:	EV= 7.0680653E+03, f= 13.390440, T= 0.074736	

Fig 9. Modal analysis results showing frequency values.

V. RESULTS AND DISCUSSION

This section presents the results obtained from the structural analysis and moving load simulations, followed by a detailed discussion of the bridge response under different loading conditions. The objective is to evaluate the dynamic behavior of the structure, identify resonance effects, and quantify their impact on stress and fatigue life.

A. Dead Load Analysis

Dead load analysis is performed to evaluate the structural response under self-weight and permanent loads, including the deck, girders, piers, and diaphragms. This analysis serves as a baseline for understanding the inherent stability and load distribution characteristics of the bridge.

The results indicate that the structure remains stable under static loading conditions, with deflection and internal force values within acceptable limits. The load is observed to be uniformly distributed across the girders, confirming the effectiveness of the structural configuration.

The dead load response establishes a reference state against which dynamic amplification under moving loads can be compared.

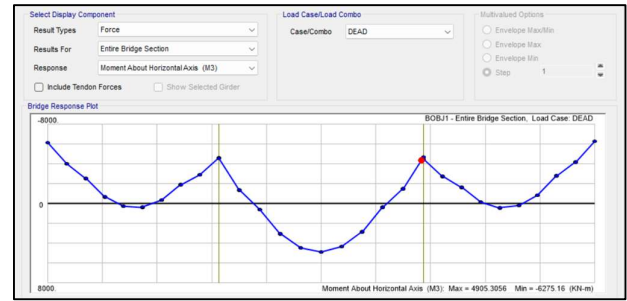


Fig 10(a). Dead load deflection diagram.

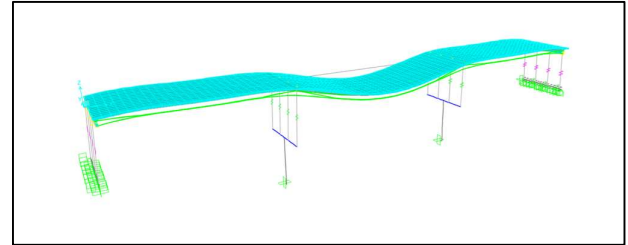


Fig 10(b). Bending deforms under dead load.

B. Prestress Analysis

Prestress analysis is carried out to evaluate the effect of pre-applied forces in the girders. The introduction of prestressing improves the load-carrying capacity of the bridge by counteracting tensile stresses and reducing overall deflection.

The results show a noticeable reduction in tensile stress zones and improved stress distribution across the girders. This contributes to enhanced durability and serviceability of the structure under operational loads.

Additionally, prestressing helps in controlling crack formation and maintaining structural integrity over long-term usage.

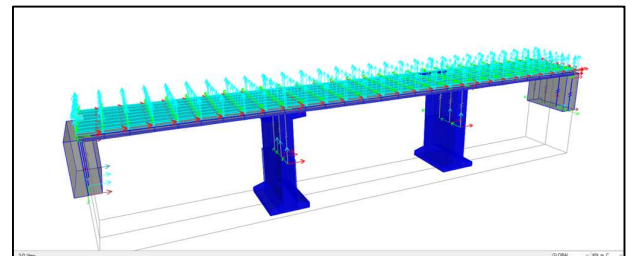


Fig 11(a). Stress distribution with prestress.

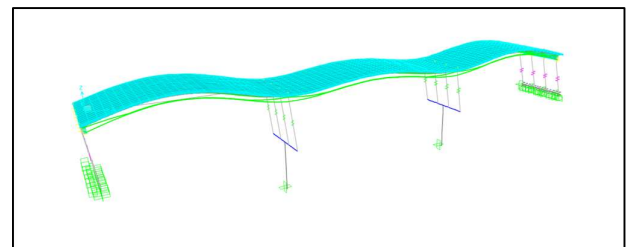


Fig 11(b). Comparison of deflection (with and without prestress).

C. Modal Analysis Results

Modal analysis is performed to determine the natural frequencies and corresponding mode shapes of the bridge. The fundamental frequency is found to be approximately **1.98 Hz**, corresponding to a dominant vibration mode.

Lower natural frequencies indicate that the structure is more susceptible to excitation from moving vehicular loads, particularly at moderate traffic speeds. Higher modes are also identified, each associated with different vibration patterns and corresponding frequencies.

These natural frequencies form the basis for identifying resonance conditions when matched with excitation frequencies induced by moving loads.

```
Original stiffness at shift : EV= 0.0000000E+00, f= .000000, T= -INFINITY-
Number of eigenvalues below shift = 0
Found mode 1 of 20: EV= 3.4313659E-07, f= 9.32E-06, T= 10726.211
Found mode 2 of 20: EV= 1.5462196E+02, f= 1.975045, T= 0.505294
Found mode 3 of 20: EV= 3.1507489E+02, f= 2.025056, T= 0.353975
Found mode 4 of 20: EV= 4.5250628E+02, f= 3.385575, T= 0.255371
Found mode 5 of 20: EV= 4.5656523E+02, f= 3.400725, T= 0.254055
Found mode 6 of 20: EV= 7.3207465E+02, f= 4.304236, T= 0.232221
Found mode 7 of 20: EV= 8.6862513E+02, f= 4.690686, T= 0.213188
Found mode 8 of 20: EV= 1.9035565E+03, f= 6.943894, T= 0.144011
Found mode 9 of 20: EV= 2.9468542E+03, f= 8.491858, T= 0.117760
Found mode 10 of 20: EV= 2.9468542E+03, f= 8.491858, T= 0.117760
Found mode 11 of 20: EV= 3.1505032E+03, f= 8.933262, T= 0.111941
Found mode 12 of 20: EV= 3.2796457E+03, f= 9.114516, T= 0.109715
Found mode 13 of 20: EV= 3.6175472E+03, f= 9.572541, T= 0.104465
Found mode 14 of 20: EV= 4.2635299E+03, f= 10.392135, T= 0.096227
Found mode 15 of 20: EV= 5.1291955E+03, f= 11.398423, T= 0.087731
Found mode 16 of 20: EV= 5.3636822E+03, f= 11.656057, T= 0.085752
Found mode 17 of 20: EV= 5.9897367E+03, f= 12.316512, T= 0.081192
Found mode 18 of 20: EV= 6.3838086E+03, f= 12.716275, T= 0.078639
Found mode 19 of 20: EV= 6.9735905E+03, f= 13.291096, T= 0.075238
Found mode 20 of 20: EV= 7.0680653E+03, f= 13.380440, T= 0.074736
```

Fig 12. Table of natural frequencies vs mode number.

D. Moving Load Analysis

Moving load analysis is conducted by simulating vehicle loads traversing the bridge at different speeds. The structural response is evaluated in terms of displacement, shear force, bending moment, and acceleration.

The results demonstrate a clear dependency of structural response on vehicle speed. As the speed varies, the excitation frequency changes, leading to fluctuations in the response parameters.

Significant peaks in displacement and internal forces are observed at specific speeds, indicating resonance conditions.

These peak responses are associated with dynamic amplification, where the bridge experiences higher stresses compared to static loading conditions. The analysis confirms that vehicle speed is a critical parameter influencing bridge performance.

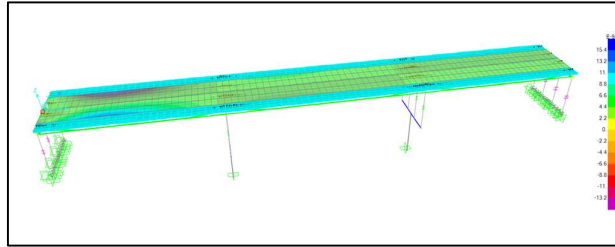


Fig 13. Bending moment variation under moving load.

E. Stress and Fatigue Analysis

The stress response of the bridge is significantly influenced by resonance effects. Under resonance conditions, the dynamic amplification leads to an increase in stress amplitude. The analysis shows that the stress increases by approximately **31.6%** compared to normal operating conditions.

This increase in stress has a direct impact on fatigue life. Since fatigue damage accumulates over repeated loading cycles, even a moderate increase in stress results in a substantial reduction in the number of cycles to failure.

Using the fatigue life relationship, it is observed that the effective fatigue life reduces to nearly **44% of its original value** under resonance conditions. This highlights the critical importance of avoiding resonance to ensure long-term structural durability.

F. Resonance Speed Ranges

Based on the modal analysis and moving load simulations, multiple resonance-prone speed ranges are identified. These speeds correspond to conditions where the excitation frequency closely matches the natural frequencies of the bridge.

The primary resonance speeds obtained from the analysis are summarized in Table III.

Mode Number	Natural Frequency (Hz)	Critical Speed (km/h)
Mode 2	1.98	28.5
Mode 3	2.83	40.6
Mode 4	3.39	48.7
Mode 5	3.40	48.9
Mode 6	4.31	62.0

Table III. Identified Resonance Speed Ranges

These speed ranges represent resonance zones that must be avoided during bridge operation. Operating within these ranges leads to peak structural response and accelerated fatigue damage.

By identifying and excluding these critical speed intervals, the proposed RASC framework enables safer and more efficient bridge operation.

VI. PROPOSED CONTROL STRATEGY

The proposed Resonance-Avoidance Speed Control (RASC) strategy is designed to mitigate resonance effects by regulating vehicle speeds instead of modifying the structural properties of the bridge. By treating traffic speed as a controllable input parameter, the framework aims to prevent excitation frequencies from coinciding with the natural frequencies of the structure. This section outlines the conceptual basis, control logic, practical implementation, and advantages of the proposed approach.

A. Concept

The central idea of the RASC framework is to control the excitation source rather than the structural response. Since the excitation frequency generated by a moving vehicle is directly dependent on its speed, regulating vehicle speed provides an effective means of avoiding resonance.

For a given bridge, natural frequencies are fixed structural properties. However, vehicle speed can be adjusted dynamically. By identifying the critical speeds corresponding to resonance conditions, it becomes possible to define *restricted speed zones* that must be avoided during operation.

Instead of allowing continuous speed variation, the framework introduces *safe operating bands*, which are ranges of vehicle speeds that do not induce resonance. Vehicles are guided to operate within these safe bands, thereby minimizing dynamic amplification effects.

B. Implementation Options

The proposed control strategy can be implemented using both conventional and advanced traffic management systems. Depending on the level of infrastructure available, the following approaches can be adopted:

- **Static Speed Regulation:** Installation of speed limit signs indicating safe speed ranges and restricted zones based on precomputed resonance conditions.
- **Dynamic Traffic Control Systems:** Use of variable message signs (VMS) and adaptive traffic signals to update speed limits in real time based on traffic conditions.
- **Sensor-Based Monitoring:** Integration of vibration sensors and structural health monitoring systems to detect real-time bridge response and adjust speed limits dynamically.
- **Intelligent Transportation Systems (ITS):** Communication between vehicles and infrastructure (V2I) to automatically regulate vehicle speeds based on resonance avoidance requirements.

These implementation options provide flexibility, allowing the framework to be applied across a wide range of bridge types and traffic environments.

VII. FUTURE WORK

While the proposed Resonance-Avoidance Speed Control (RASC) framework demonstrates effective mitigation of resonance through speed regulation, several extensions can further enhance its applicability, accuracy, and real-world implementation.

One important direction is the integration of **real-time structural health monitoring systems**. By incorporating vibration sensors, accelerometers, and data acquisition systems, the bridge response can be continuously monitored. This would enable dynamic identification of resonance conditions under varying traffic and environmental influences, allowing the control strategy to adapt in real time rather than relying solely on precomputed values.

Another area of improvement lies in the development of **adaptive and intelligent control algorithms**. Machine learning and data-driven techniques can be employed to predict resonance conditions based on traffic patterns, vehicle types, and environmental factors. Such approaches can optimize speed regulation more efficiently and improve system responsiveness under complex, real-world conditions.

The current study is based on a simplified moving load model. Future work can extend the analysis to include **multi-vehicle traffic scenarios**, where multiple vehicles with varying axle loads, spacing, and speeds interact simultaneously with the bridge. This would provide a more realistic representation of traffic-induced excitation and its impact on structural response.

Additionally, the methodology can be applied to **different bridge types**, such as cable-stayed and suspension bridges, to evaluate its effectiveness across varying structural configurations and dynamic characteristics.

Further investigation can also consider the influence of **environmental factors**, including temperature variations, wind loads, and material nonlinearities, which may affect the dynamic properties of the bridge over time.

Finally, **experimental validation** of the proposed framework is essential. Scaled physical models or field testing on existing bridges can be used to verify the accuracy of simulation results and assess the practical feasibility of implementing the RASC strategy in real-world conditions.

VIII. CONCLUSION

This study presents a simulation-based investigation of resonance effects in bridge structures subjected to moving vehicular loads and proposes a Resonance-Avoidance Speed Control (RASC) framework as an effective mitigation strategy. The analysis demonstrates that resonance occurs when the excitation frequency induced by vehicle speed approaches the natural frequency of the bridge, resulting in significant amplification of structural response.

Through modal and moving load analyses, critical resonance speed ranges were identified for the selected continuous girder bridge. The results show that operating within these speed ranges leads to increased stress levels and accelerated fatigue damage. Quantitatively, the stress was observed to increase by approximately 31.6% under resonance conditions, leading to a reduction in fatigue life to nearly 44% of its original value.

To address this issue, the proposed RASC framework introduces a non-intrusive control approach by regulating vehicle speeds to avoid resonance-prone regions. By defining safe operating speed bands and restricting critical speed intervals, the framework effectively reduces dynamic amplification without requiring structural modifications.

Overall, the findings highlight that controlling traffic speed can serve as a practical and cost-effective solution for mitigating resonance in bridges. The proposed approach not only improves structural performance and extends service life but also offers strong potential for integration with modern traffic management and smart infrastructure systems.

IX. REFERENCES

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